

Glass Manufacturing

Glass batch chemistry with high-purity aragonite

Furnace efficiency, batch homogeneity and CO₂ accounting for modern glass formulations.

Summary

Calcium carbonate is a primary CaO source in most glass batch formulations. The purity, particle size distribution and iron content of the CaCO₃ input directly influence melt rate, batch homogeneity, optical clarity and furnace energy consumption. High-purity natural oolitic aragonite offers a fine, uniformly classified, low-iron alternative to conventional limestone for container, flat, solar, fiberglass, specialty and optical glass producers.

Why aragonite in glass batches

Higher CaO contribution per ton

Designed to support efficient CaO delivery in glass batches via consistent CaCO₃ chemistry.

Furnace efficiency

Fine, uniform oolitic particles can help evaluate improvements in melt rate and batch homogeneity.

Lower-iron, optical-grade purity

High whiteness and low impurity profile suited for clarity-critical glass.

CO₂ emissions opportunity

Ocean-renewed source supports sustainability accounting for batch-related carbon.

Glass-grade specification snapshot

Typical chemistry (raw material)

Constituent	Typical	Notes
CaCO ₃	≥ 97.5%	Calcium carbonate, primary
MgCO ₃	≤ 0.9%	Magnesium carbonate
SiO ₂	≤ 0.6%	Silica
Fe ₂ O ₃	≤ 0.04%	Iron oxide — low
Al ₂ O ₃	≤ 0.10%	Alumina
Na ₂ O + K ₂ O	≤ 0.30%	Alkalis
LOI	~ 43%	Loss on ignition

Applications

Container glass, flat glass, solar glass, fiberglass, specialty glass and optical glass formulations. Application-specific batch performance depends on furnace design, cullet ratio and full batch formulation — AragoCor's applications engineering team supports documented batch trials to validate melt rate, homogeneity and clarity outcomes before full specification.

Conclusion

For glass producers evaluating CaCO_3 sources on technical merit, aragonite's fine oolitic morphology, high whiteness and low iron content make it a candidate worth a documented batch trial — particularly where furnace efficiency and optical clarity are competitive differentiators.

This brief summarizes typical properties of AragoCor natural oolitic aragonite for comparative and educational purposes. Values are representative; request a current certificate of analysis for lot-specific data. © AragoCor Minerals LLC. info@aragocorminerals.com · +1 (855) 751-9100